

IOWA STATE UNIVERSITY  
ELECTRICAL AND COMPUTER ENGINEERING

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EE 2300

# Lab 6 Report – Comparators & a non-linear oscillator

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Date of Experiment: March 24, 2026

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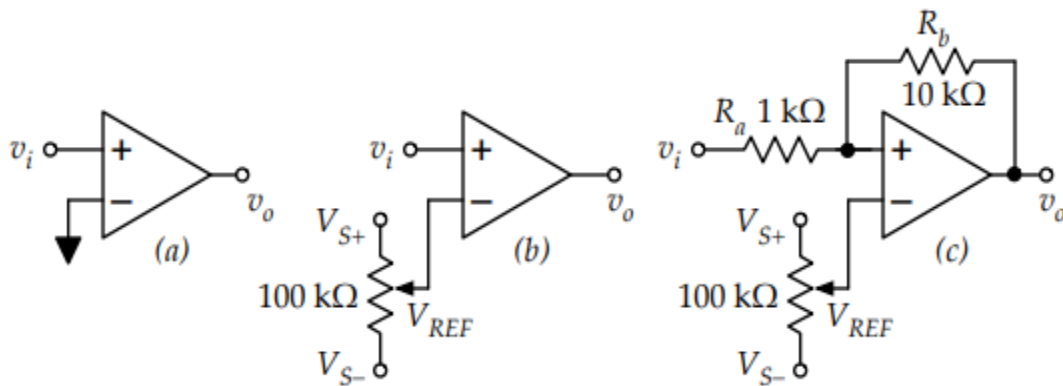
Name of Group Members: Sayon Saha, Aaron Kammlah

**Sayon Saha**

## Introduction

In this lab, we started working with comparators and non-linear oscillators. In experiment 1, we built a non-inverting comparator in stages, and each time analyzed the input and output voltage on the oscillator both in normal and XY mode. We also calculated the higher threshold and lower threshold voltages of the circuit in part c. For experiment 3, we built an inverting comparator and did the same analysis as experiment 1. For experiment 3, we built a non-linear oscillator first without a potentiometer and then with one. We monitored the input and output voltage both times. We also calculated the expected higher threshold and lower threshold voltage along with the oscillation frequency.

### Experiment 1. Non-inverting Comparator



|                             |                        |
|-----------------------------|------------------------|
| Calculated $V_{TL}$ : 3 V   | Measured value: 3 V    |
| Calculated $V_{TH}$ : 1.4 V | Measured value: 1.37 V |

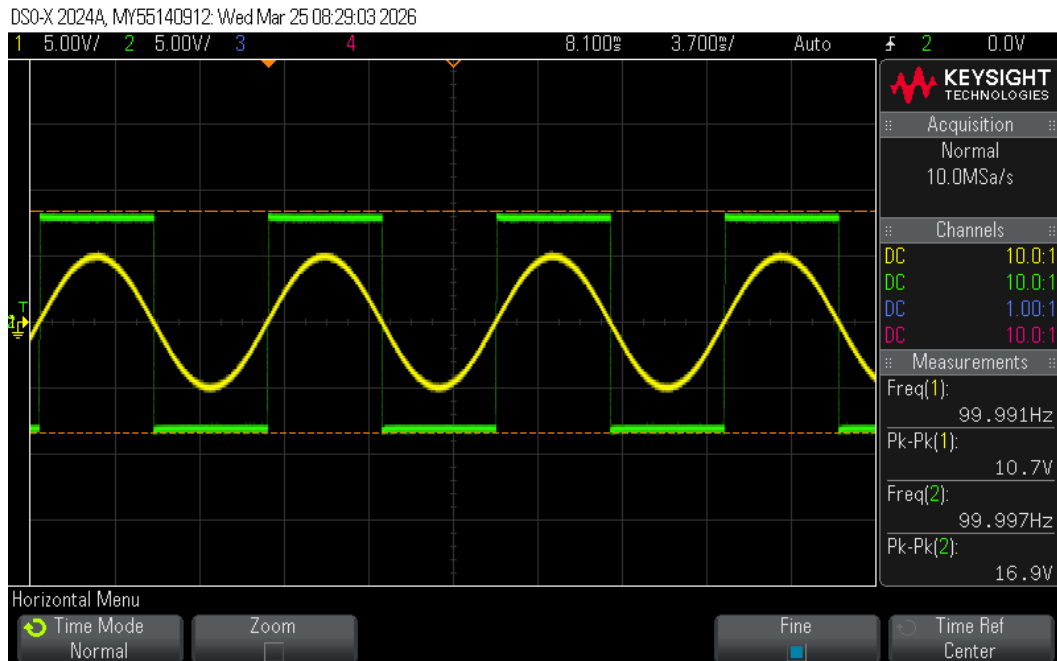


Figure 1-1. Plot of input and output for a

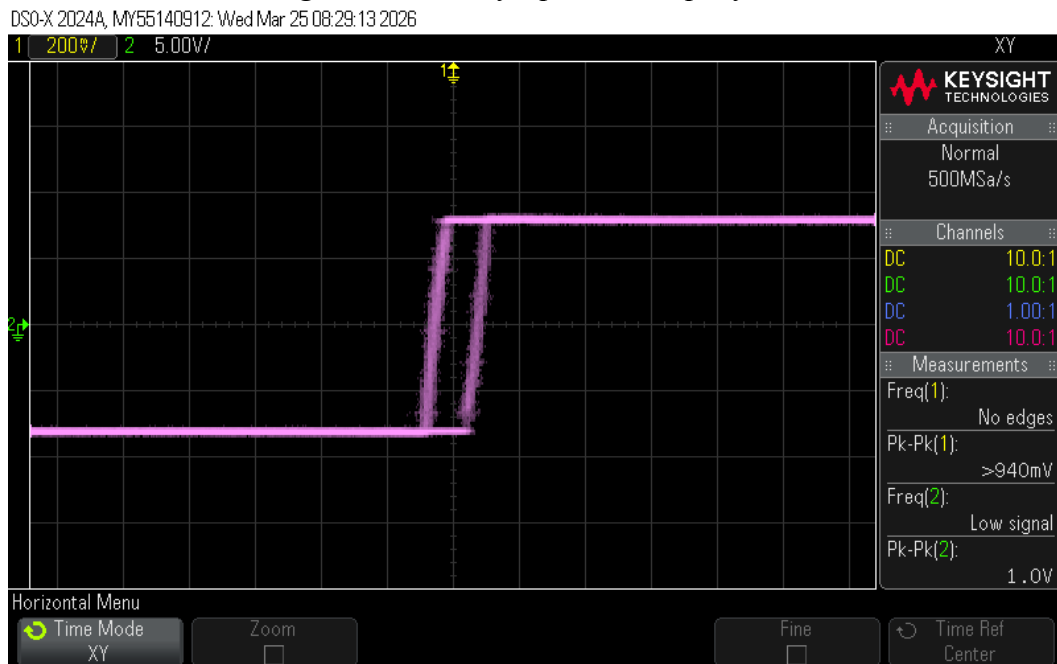


Figure 1-2. Plot of XY mode for a

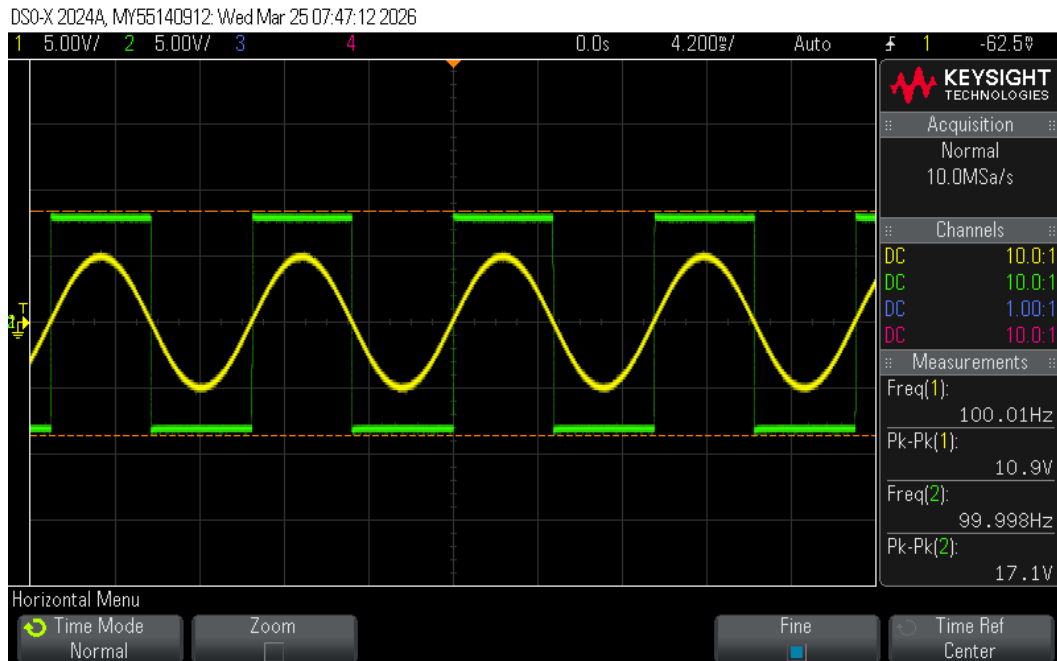


Figure 1-3. Plot of input and output for  $b$

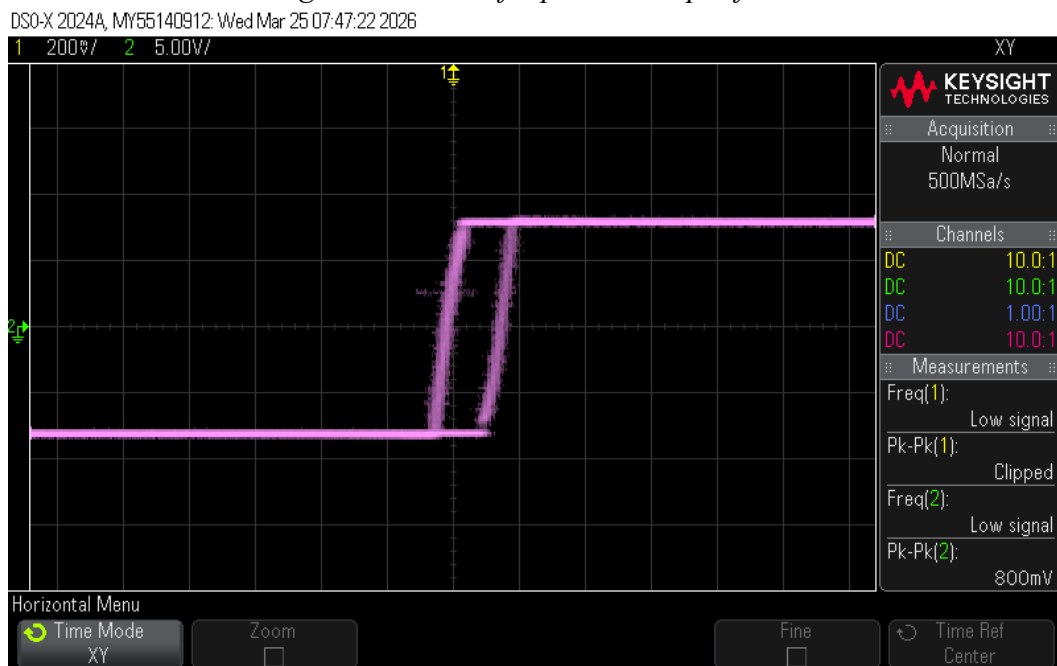


Figure 1-4. Plot of XY mode for  $b$

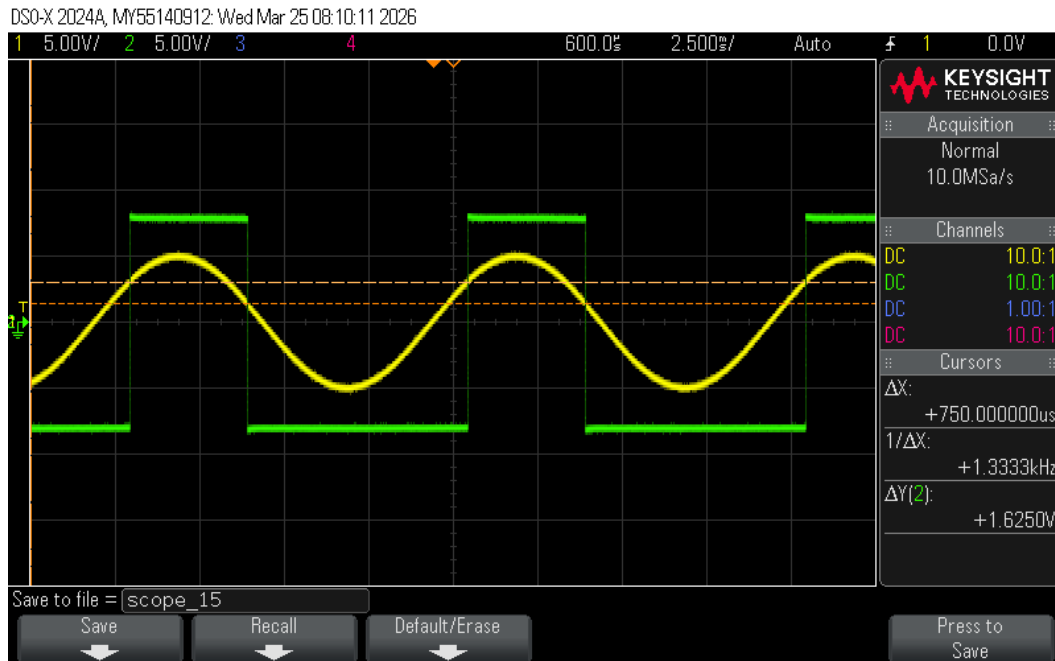


Figure 1-5. Plot of input and output for  $c$

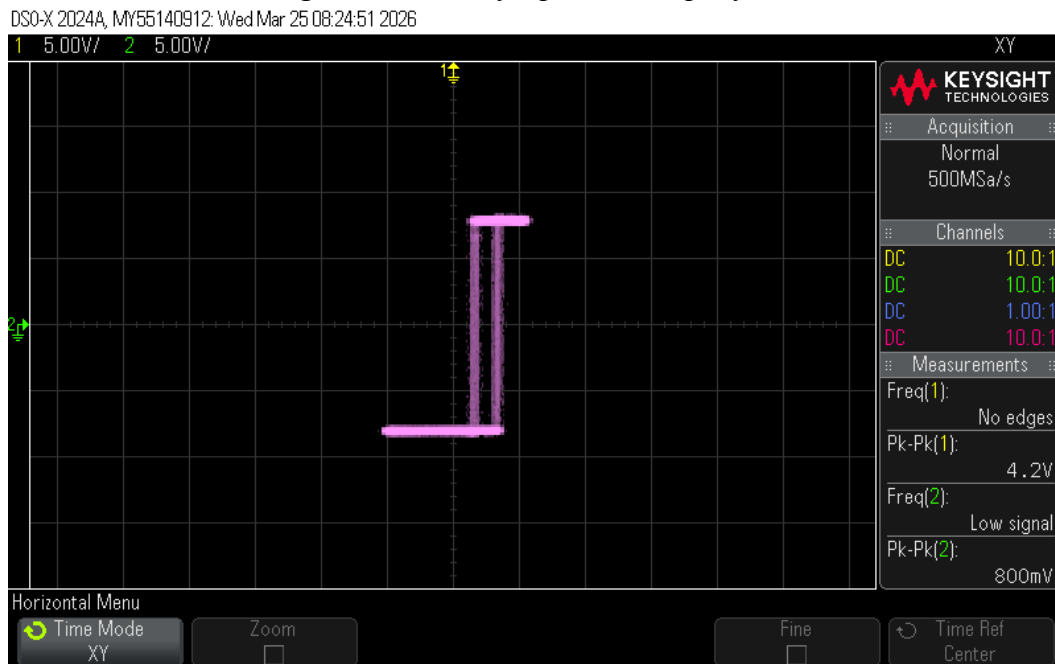
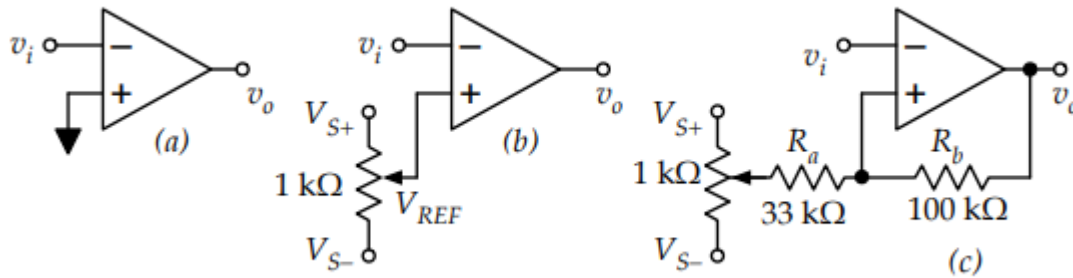


Figure 1-6. Plot of XY mode for  $c$

The part a and b traces for this experiment are almost identical, but a noticeable shift can be seen in part c when added the feedback loop, especially in XY mode. The calculated and measured values are also almost identical.

## Experiment 2. Inverting Comparator



Calculated  $V_{TL}$ : 3.49 V Measured value: 3.68 V

Calculated  $V_{TH}$ : -0.48 V Measured value: 0.312 V

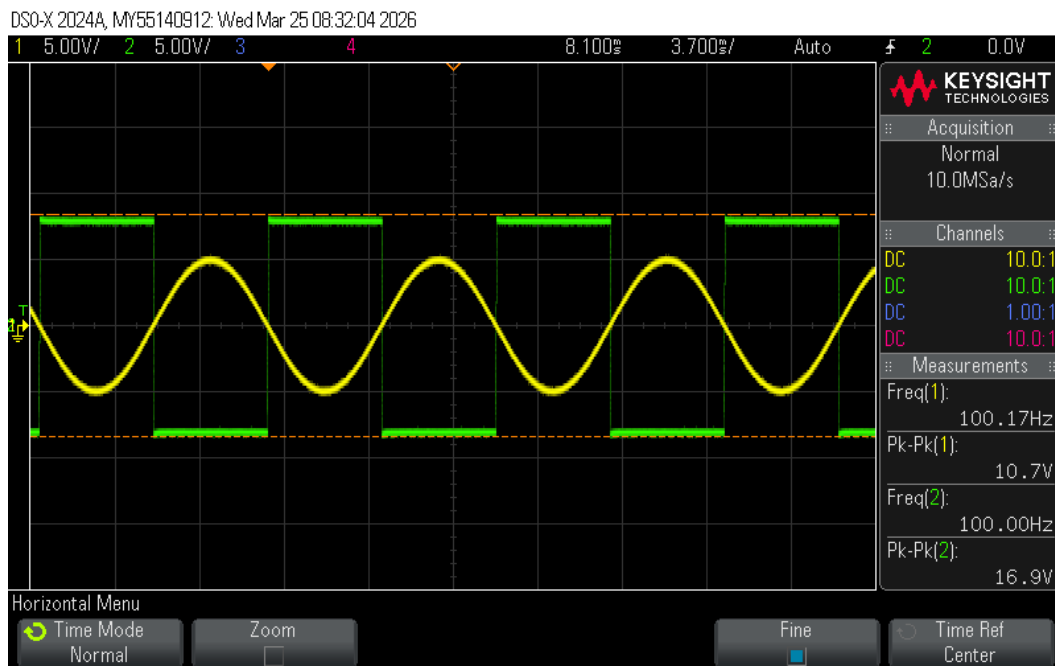


Figure 2-1. Plot of input and output for a

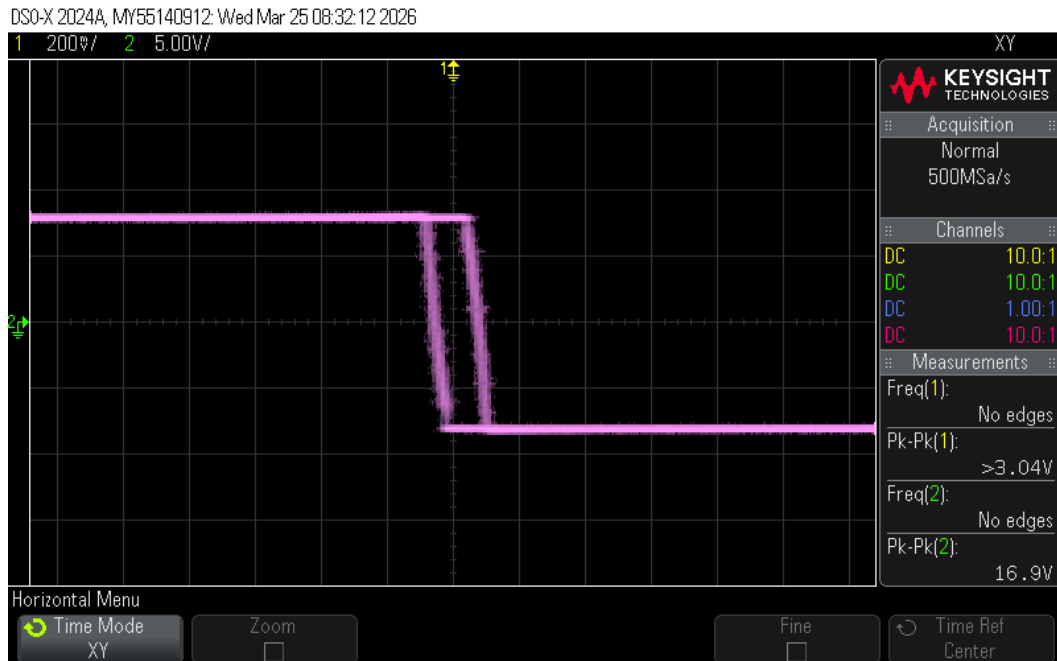


Figure 2-2. Plot of XY mode for a

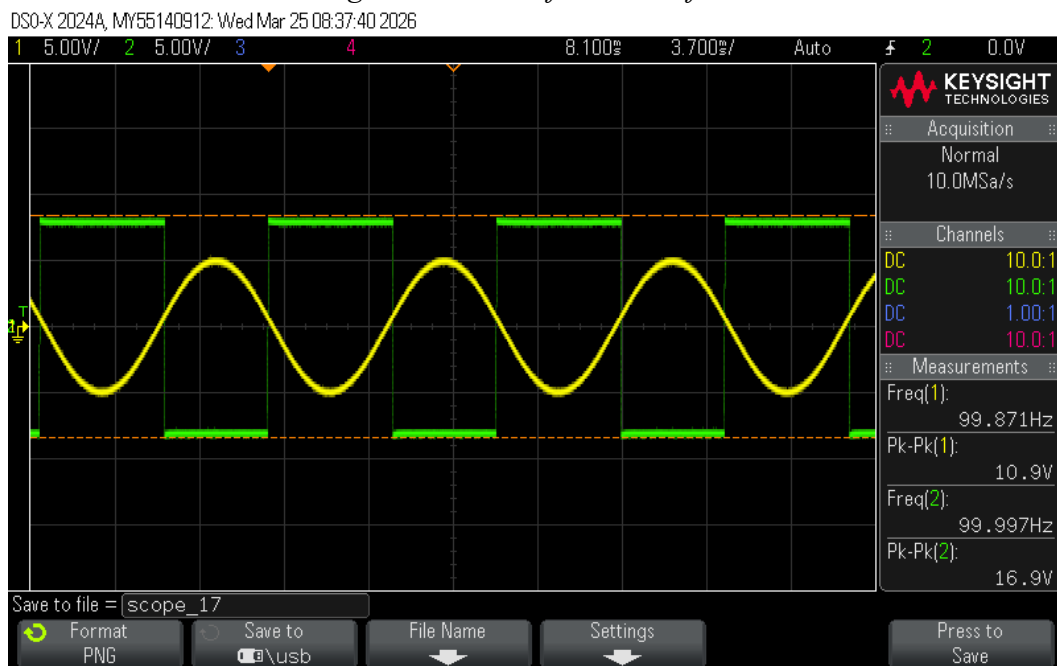


Figure 2-3. Plot of input and output for b

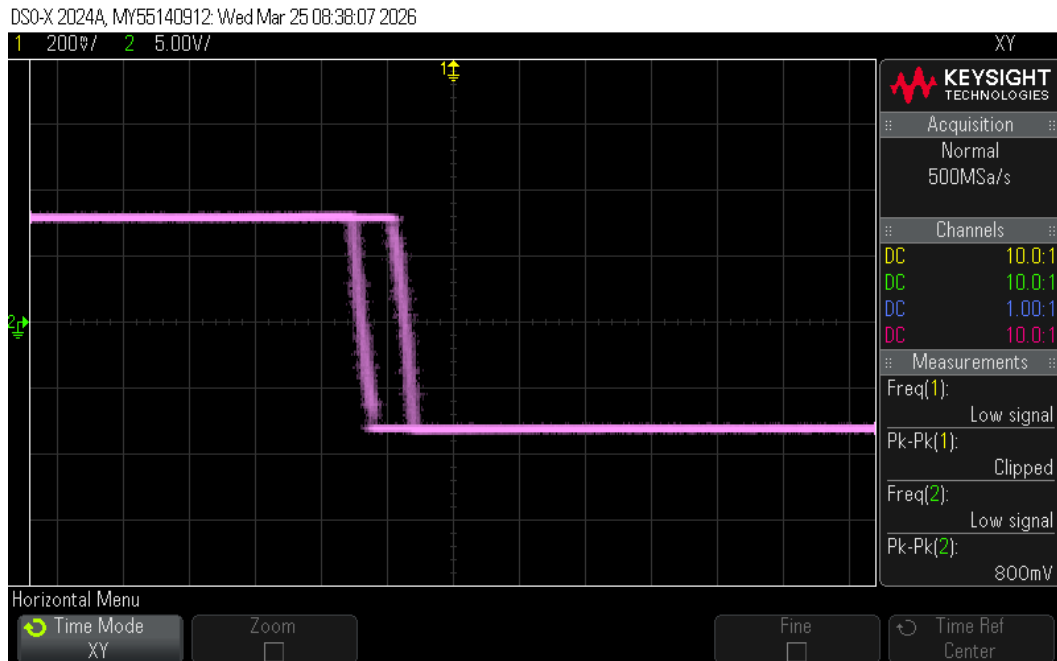


Figure 2-4. Plot of XY mode for  $b$

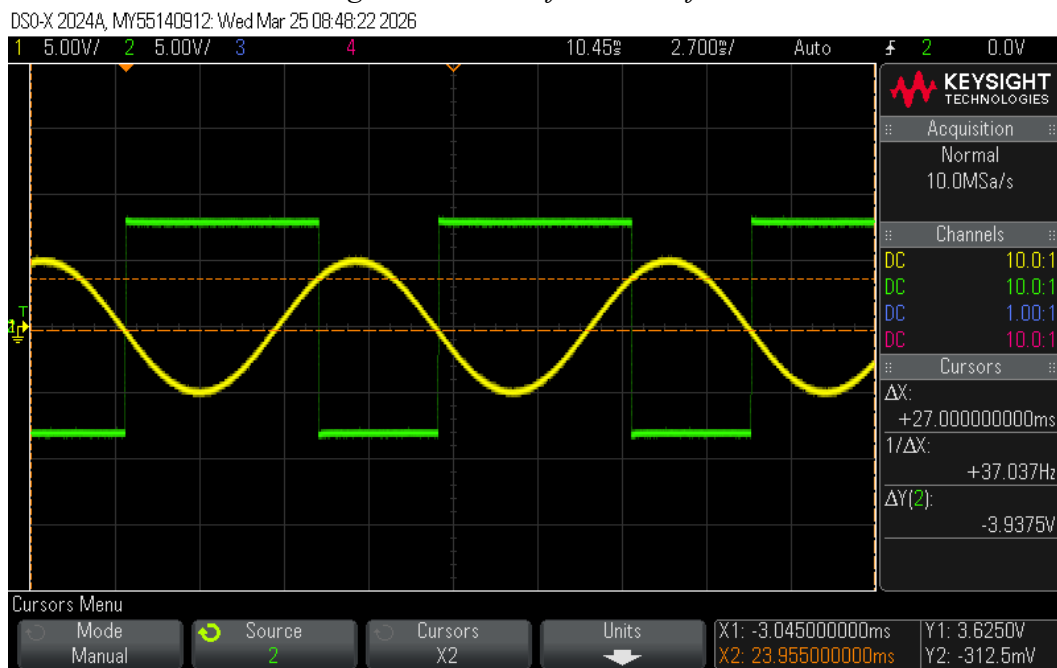


Figure 2-5. Plot of input and output for  $c$

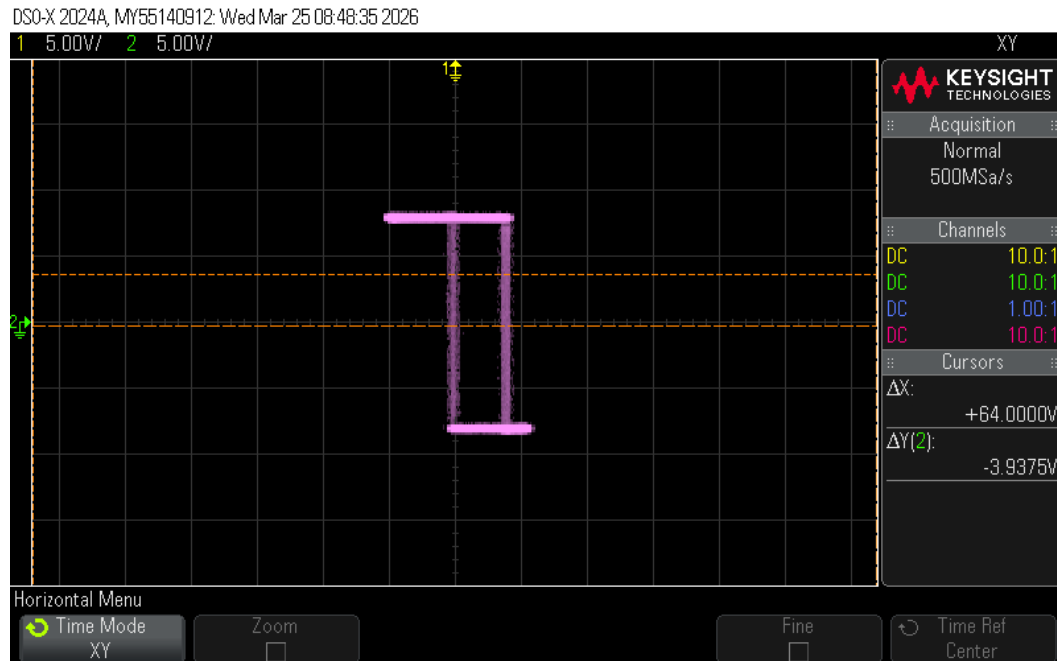
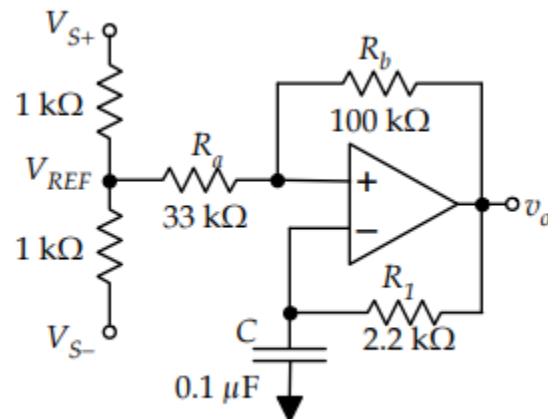


Figure 2-6. Plot of XY mode for c

Same as experiment 1, in part a and part b of this experiment, we find almost identical traces. But in part c, we see a noticeable change when we added in the feedback loop. The calculated and measured values for higher threshold and lower threshold voltages are almost the same too.

## Experiment 3. Non-linear Oscillator



Calculated  $V_{TH}$ : 1.98 V    Measured value: 2 V

Calculated  $V_{TL}$ : -1.98 V    Measured value: -2.1 V

Calculated  $f_{osc}$ : 4.48 kHz    Measured value: 5.2 kHz

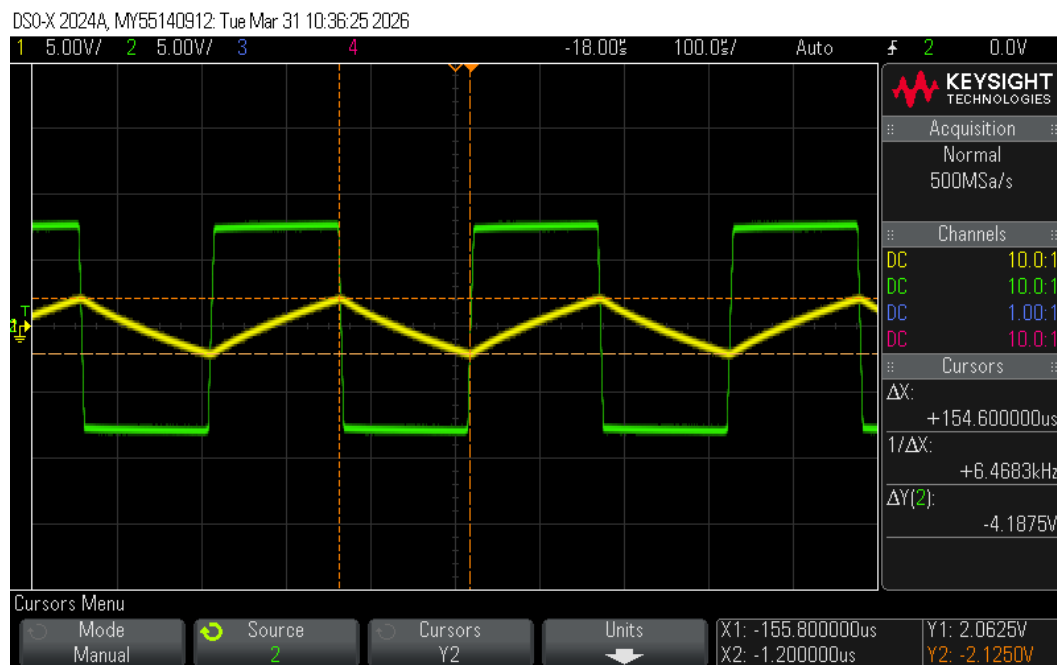
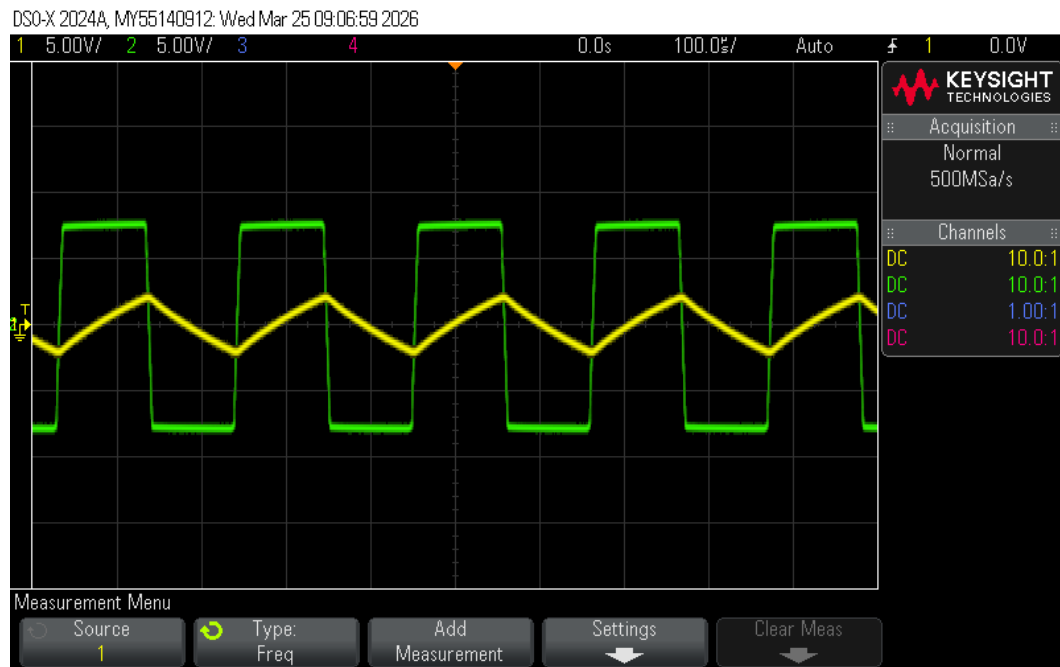


Figure 1-1. Plot of input and output without potentiometer



*Figure 1-1. Plot of input and output with potentiometer*

The calculated and measured values were close. The calculated and measured higher thresholds and lower thresholds were within 5% of each other. The oscillation frequency was also close to the calculated one.

## **Conclusion**

In this lab we didn't face any major problems. Our initial measurement of  $V_{TH}$  and  $V_{TL}$  were wrong because of cursor misplacement. They didn't agree with our calculations. That issue was addressed when we redid the experiment and placed the cursors where they needed to be. Other than that, the experiments were completed without any complications.

## Calculation

### Experiment 01:

from the circuit,

$$V_f = V_- = V_{REF}$$

at the non-inverting node,

$$\frac{V_f - V_i}{1000} + \frac{V_f - V_o}{10000} = 0$$

$$\Rightarrow \frac{V_{REF} - V_i}{1000} + \frac{V_{REF} - V_o}{10000} = 0$$

$$\Rightarrow V_{REF} \left( \frac{1}{1000} + \frac{1}{10000} \right) = \frac{V_i}{1000} + \frac{V_o}{10000}$$

$$\Rightarrow V_i = V_{REF} \left( 1 + \frac{1}{10} \right) - V_o \frac{1}{10}$$

assuming  $V_{REF} = 2V$ ,

$$V_i = 2(1.1) - 0.1V_o$$

$$\Rightarrow V_i = 2.2 - 0.1V_o$$

now,

upper threshold happen when the input is low,

$$\text{so, } V_{iTH} = 2.2 - 0.1(-8)$$

$$\Rightarrow V_{iTH} = 3V$$

lower threshold happens when the input is high,

$$\text{so } V_{TL} = 2.2 - 0.1(8)$$

$$\Rightarrow V_{TL} = 1.4 \text{ V}$$

so, upper threshold,  $V_{TH} = 3 \text{ V}$

lower threshold,  $V_{TL} = 1.4 \text{ V}$

hysteresis,  $V_H = V_{TH} - V_{TL} = 1.6 \text{ V}$

### Experiment 02:

From the circuit, switching happens when

$$V_+ = V_- = V_i$$

at the non-inverting node,

$$\frac{V_+ - V_{REF}}{33\text{K}} + \frac{V_+ - V_0}{100\text{K}} = 0$$

$$\Rightarrow \frac{V_i - V_{REF}}{33\text{K}} + \frac{V_i - V_0}{100\text{K}} = 0$$

$$\Rightarrow V_i \left( \frac{1}{33\text{K}} + \frac{1}{100\text{K}} \right) = \frac{V_{REF}}{33\text{K}} + \frac{V_0}{100\text{K}}$$

$$\Rightarrow V_i = \frac{V_{REF}/33\text{K} + V_0/100\text{K}}{1/33\text{K} + 1/100\text{K}}$$

Higher threshold is reached when output is high,

$$\text{so, } V_{TH} = \frac{2/33k + 8/100k}{1/33k + 1/100k}$$

$$\Rightarrow V_{TH} = 3.49 \text{ V}$$

Lower threshold is reached when output is low,

$$\text{so, } V_{TL} = \frac{2/33k - 8/100k}{1/33k + 1/100k}$$

$$\Rightarrow V_{TL} = -0.48 \text{ V}$$

so, Higher threshold,  $V_{TH} = 3.49 \text{ V}$

Lower threshold,  $V_{TL} = -0.48 \text{ V}$

Hysteresis,  $V_H = V_{TH} - V_{TL}$

$$= 3.49 - (-0.48)$$

$$= 3.97 \text{ V}$$

### Experiment 03:

From the circuit,

$$V_{REF} = 0$$

at the non-inverting node,

$$\frac{V_+ - V_{REF}}{33k} + \frac{V_+ - V_o}{100k} = 0$$

$$\Rightarrow V_+ \left( \frac{1}{33k} + \frac{1}{100k} \right) = \frac{V_o}{100k}$$

$$\Rightarrow V_+ = \frac{V_o / 100k}{(1/33k + 1/100k)}$$

$$\Rightarrow V_i = V_o \frac{1}{1 + 100k/33k}$$

$$\Rightarrow V_i = 0.248 V_o$$

So,  $V_{TH}$  happens at  $V_o = 8V$

$$V_{TH} = 0.248 \times 8 = 1.98V$$

$V_{TL}$  happens at  $V_o = -8V$

$$V_{TL} = 0.248 \times (-8) = -1.98V$$

We know, for a relaxation oscillation circuit,

$$\omega_{osc} = \frac{1}{R_1 C \ln\left(\frac{1+\beta}{1-\beta}\right)} \quad \text{where } \beta = \frac{R_a}{R_a + R_b} = 0.248$$

so, oscillation frequency,

$$\begin{aligned} f_{osc} &= \frac{1}{2 R_1 C \ln\left(\frac{1+\beta}{1-\beta}\right)} \\ &= 4486.6 \text{ Hz} \\ &\approx 4.48 \text{ kHz} \end{aligned}$$

$$\text{so, } V_{TH} = 1.8 \text{ V}$$

$$V_{TL} = -1.8 \text{ V}$$

$$f_{osc} = 4.48 \text{ kHz}$$